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GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 3

Class Test

COURSE NAME: Deep Learning **COURSE CODE: MLA0403**
Faculty: Dr. Arul Raj A.M **Slot : A**

COURSE OUTCOME COVERED

CO 2: Analyse Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

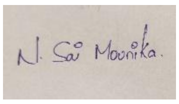
Assessment Tool Weightage: 20%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I Narra Sai Mounika Reg no: 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Narra Sai Mounika
Name of the student:

Class Test

CO2-AT3

QUESTIONS: 05 **Total Marks:25**

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

Answer:

Aspect	**Machine Learning (ML)**	**Deep Learning (DL)**
Architecture	Uses algorithms such as Decision Trees, SVM, and Random Forest	Uses multi-layer Artificial Neural Networks
Data Requirements	Works well with small and medium datasets	Requires large amounts of data
Feature Extraction	Manual feature engineering is required	Automatically learns features from data
Computational Power	Lower computational requirements	Requires high computational power (GPU/TPU)
Performance	Effective for structured data	Excellent for unstructured data such as images, audio, and text

Machine Learning Applications:

1. Email Spam Detection
2. Credit Card Fraud Detection

Deep Learning Applications:

1. Facial Recognition Systems
2. Self-Driving Cars

Conclusion:

Machine Learning relies on handcrafted features, whereas Deep Learning automatically extracts features and performs better on complex datasets.

2. Explain the concept of **Representation Learning**. Discuss how the **width** and **depth** of a neural network influence its learning capability, computational complexity and model performance.

Answer:

Representation Learning is the ability of a model to automatically learn meaningful features from raw data without manual feature engineering.

Examples:

- Learning facial features from images.
- Learning customer behaviour patterns from purchase history.

Influence of Width

Width refers to the number of neurons in a hidden layer.

Advantages:

- Learns more patterns simultaneously.
- Improves feature representation.

Disadvantages:

- Higher memory usage.
- Increased risk of overfitting.

Influence of Depth

Depth refers to the number of hidden layers.

Advantages:

- Learns hierarchical features.
- Captures complex relationships in data.
- Improves accuracy for difficult tasks.

Disadvantages:

- Higher computational complexity.
- Longer training time.
- Possibility of vanishing gradients.

Impact on Performance

Factor	**Wide Network**	**Deep Network**
Learning Capability	Moderate	High
Computational Cost	Medium	High
Feature Learning	Limited	Hierarchical

Conclusion:

A balanced combination of width and depth helps achieve better performance and generalization.

3. Compare **ReLU**, **Leaky ReLU (LReLU)** and **ELU (Exponential Linear Unit)** activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.

Answer:

Activation Function	**Formula**
ReLU	$f(x)=\max(0,x)$
Leaky ReLU	$f(x)=x$ if $x>0$, αx if $x\leq 0$
ELU	$f(x)=x$ if $x>0$, $\alpha(e^{x}-1)$ if $x\leq 0$

ReLU (Rectified Linear Unit)

Advantages:

- Simple and computationally efficient.
- Faster convergence.
- Reduces vanishing gradient problem.

Disadvantages:

- Suffers from "Dead Neuron" problem.

Use Cases:

- CNNs
- Image Classification

Leaky ReLU (LReLU)

Advantages:

- Solves dead neuron problem.
- Allows small gradients for negative inputs.

Disadvantages:

- Requires tuning of α parameter.

Use Cases:

- Deep Neural Networks
- Object Detection Models

ELU (Exponential Linear Unit)

Advantages:

- Produces smoother learning.
- Negative outputs help maintain zero-centered activations.

Disadvantages:

- Computationally more expensive.

Use Cases:

- Deep Networks requiring stable convergence.

Comparison

Feature	**ReLU**	**Leaky ReLU**	**ELU**
Speed	Fast	Fast	Moderate
Dead Neurons	Yes	No	No
Computational Cost	Low	Low	High
Accuracy	Good	Better	Best in some cases

4. Explain the concept of **unsupervised learning** in neural networks. Describe the working principle of **Restricted Boltzmann Machines (RBMs)** and **Autoencoders** and compare their objectives and applications.

Answer:

Unsupervised Learning

Unsupervised learning trains models using unlabeled data to discover hidden patterns, structures, and relationships.

Applications:

- Clustering
- Recommendation Systems

- Anomaly Detection

Restricted Boltzmann Machines (RBMs)

RBMs are generative neural networks consisting of:

1. Visible Layer (Input Data)
2. Hidden Layer (Feature Extraction)

Working Principle:

- Input data is fed into visible units.
- Hidden units learn latent patterns.
- The network reconstructs the input.
- Weights are updated to minimize reconstruction error.

Applications:

- Movie Recommendation Systems
- Collaborative Filtering
- Feature Learning

Autoencoders

Autoencoders are neural networks used for learning compressed data representations.

Components:

1. Encoder
2. Bottleneck Layer
3. Decoder

Working Principle:

- Encoder compresses data.
- Bottleneck stores essential information.
- Decoder reconstructs original data.

Applications:

- Dimensionality Reduction
- Noise Removal
- Anomaly Detection

Comparison

Feature	RBM	Autoencoder
Type	Generative Model	Neural Network
Learning Method	Probabilistic	Reconstruction-Based
Objective	Learn data distribution	Learn compressed representation
Applications	Recommendation Systems	Feature Extraction, Compression

Conclusion:
RBMs focus on learning probability distributions, while Autoencoders focus on efficient feature representation.

5. Explain the concept of Autoencoder and its uses in Deep learning.

Answer: An Autoencoder is an unsupervised neural network that learns to encode input data into a compressed representation and then reconstruct it.

Architecture

1. Input Layer
2. Encoder
3. Bottleneck Layer
4. Decoder
5. Output Layer

Working Process

1. Input data is provided to the encoder.
2. Encoder compresses the data into lower-dimensional features.
3. Bottleneck stores the most important information.
4. Decoder reconstructs the original data.
5. Reconstruction error is minimized during training.

Advantages

- Automatic feature learning.
- Noise reduction.
- Data compression.
- Efficient representation learning.

Uses of Autoencoders

1. Dimensionality Reduction
    - Reduces high-dimensional datasets.
2. Anomaly Detection
    - Detects abnormal patterns in ECG, IoT, and sensor data.

3. Image Denoising
    - Removes noise from images.
4. Feature Extraction
    - Generates useful features for classification tasks.
5. Data Compression
    - Reduces storage requirements.
6. Recommendation Systems
    - Learns user preferences and behavior patterns.

Conclusion:
Autoencoders are powerful deep learning models used for representation learning, dimensionality reduction, anomaly detection, and data compression, making them valuable in many real-world applications**.**

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand